



**DR. MUHAMMED ARIF**

**HOME ADDRESS**

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**OFFICE ADDRESS**

DEPARTMENT: PHARMACEUTICAL CHEMISTRY  
FACULTY OF PHARMACY,  
UNIVERSITY OF KARACHI  
KARACHI-75270, PAKISTAN

PROFESSIONAL STATUS:

PROFESSOR.

FIELD OF STUDY

PHARMACEUTICAL SCIENCES

FIELD OF SPECIALIZATION

MEDICINAL CHEMISTRY and DRUG DESIGN

EDUCATION

Ph.D. IN PHARMACEUTICAL CHEMISTRY

**PROFESSIONAL EXPERIENCE**

|                        |                                                                               |                            |                                             |
|------------------------|-------------------------------------------------------------------------------|----------------------------|---------------------------------------------|
| 1. Professor           | Pharmaceutical                                                                | Jan. 2001 to date          | Teaching & Research                         |
| 2. Chairman            | Pharmaceutical<br>Chemistry,<br>Faculty of Pharmacy,<br>University of Karachi | 2004 - 2006                | Teaching, Research &<br>Administrative work |
| 3. Associate Professor | Pharmaceutical<br>Chemistry,<br>University of Karachi                         | Sept. 1996<br>to Dec. 2000 | Teaching & Research                         |

|                                            |                                                     |                            |                                              |
|--------------------------------------------|-----------------------------------------------------|----------------------------|----------------------------------------------|
| 4. Assistant Professor                     | Pharmaceutical Chemistry,<br>University of Karachi. | Dec. 1990<br>to Sept. 1996 | Teaching & Research                          |
| 5. Assistant Professor<br>(F. S. L)        | Philadelphia University,<br>Jordan.                 | Feb. 1995<br>to Sept. 1996 | Teaching                                     |
| 6. Visiting Assistant Professor (on F.S.L) | Jamia Al Isra Jordan                                | Session 1995-96            | Teaching                                     |
| 7. Lecturer                                | Pharmaceutical Chemistry<br>University of Karachi   | June 1988<br>to Dec. 1990  | Teaching & Research                          |
| 8. Full Time Cooperative Teacher           | Pharmaceutical Chemistry,<br>University of Karachi. | Dec. 1987<br>to June. 1988 | Teaching & Research                          |
| 9. University Research Fellow              | Pharmaceutical Chemistry,<br>University of Karachi. | July 1985<br>to Sept. 1987 | Research & Teaching                          |
| 10. Research Scholar                       | Pharmaceutical Chemistry,                           | June 1981<br>to July 1985  | Research & Teaching<br>University of Karachi |

#### **OTHER EXPERIENCES**

- Nominated as a Member Committee, Constituted by the Vice Chancellor University of Karachi for Members of Academic Council on the Board of Governors for K.M.A. Degree Girls College, Karachi.
- Member Board of Studies, Department of Pharmaceutical Chemistry.
- Member Board of Studies, Faculty of Pharmacy, University of Karachi.
- Member Senate, University of Karachi.
- Member Academic Council, University of Karachi.
- Internal Tabulator, Faculty of Pharmacy, University of Karachi.
- Member Organizing Committee and Coordinator "First International Conference on Pharmaceutical Sciences".

- Member Organizing Committee and Coordinator "Second International Conference on Pharmaceutical Sciences".
- Appointed as internal Examiner for Ph.D. thesis in Pharmaceutical Chemistry.
- Appointed as External Examiner for M. Sc. Project in Chemistry.
- Nominated as a Member Committee, Constituted by the Vice Chancellor University of Karachi for the introduction of Bachelor's of Homeopathy Degree and to formulate the Rules, Regulation and Syllabi for the same.
- Nominated as Academic Advisor, Faculty of Pharmacy, University of Karachi.
- Member library committee, University of Karachi.
- Appointed as external examiner for B. Pharm. MBA Exam, Hamdard University, Karachi.
- Member of a committee constituted by Health Department, Government of Sind to recommend syllabi for the degree of Bachelor of Homeo Medicine and Surgery and other related requirements for recognition of Homeopathy for medical degree.
- Member of the Academic Committee of the Board of Intermediate Education, Karachi.
- Administrative work of the department, faculty and University, when and where required

## **RESEARCH AND GUIDANCE**

Since the completion of Master's degree I am engaged in teaching and research. During this period I have gained experience in synthesis of biologically active compounds and evaluation of their pharmacological properties. Up-till now, two Ph.D and one M.Pharm/M.Phil degrees have been produced and more students are pursuing their research work under my supervision.

The results of above research activities have yielded in more than 35 research papers and 5 articles of general interest published in national and international journals.

## **CURRENT RESPONSIBILITIES**

- Teaching following courses to Graduate and Postgraduate Students.
  1. Medicinal Chemistry
  2. Drug design
  3. Macromolecular Targets for Drug Designing.
  4. Organic Pharmaceutical Chemistry
  5. Organic Pharmaceutical Preparations (Lab. Course)
  6. Inorganic Pharmaceutical Chemistry
- Supervision and guidance of research students for M.Phil/Ph. D. Degree.

## **PERSONAL**

**Date of birth, 29<sup>th</sup> November 1953, married with two children**

## **MEMBERSHIP OF LEARNED SOCIETIES**

1. Member Scientific Society of Pakistan.
2. Life member Pakistan Pharmacological Society.
3. Member Pakistan Biochemical Society.
4. Member Chemical Society of Pakistan.
5. Life Member Pakistan Association of Scientists and Scientific Professionals.
6. Member International Study Group for Biochemical and Biophysical Research in Pakistan (ISBBP).

## **PUBLICATIONS/PRESENTATION.**

More than 35 research papers presented / published/accepted for publication in National and International Conferences / Symposium and Journals.

Five articles of general interests have also been published.

## **RESEARCH PUBLICATIONS**

1. Effect of Aspirin and Nicotinic acid Derivatives as Potential Hypocholesterolemic Agents. Z. S. Saify, M. Arif, S.A. Ali and S. Rashid. *Pakistan Heart Journal*, 18 (3); 48-51, July 1985
2. In Vitro studies of Modified Thiosemi Carbazone Compounds for Antibacterial, Antiviral and Genetic activity. S. Ahmed, S. A. Rasool, R. Mushtaq, Z. S. Saify and M. Arif, *J. Science*, University of Karachi 14 (1&2), 47-52, Dec. 1986.
3. Effect of Aspirin and Nicotinic acid Derivatives on hyperlipoproteinemic Rabbit. Z. S. Saify, S. A. Ali, M. Arif and S. Rashid. *Pakistan Heart Journal*, 20 (2):2-5, Jan. 1987.
4. Some Thiosemicarbazones of Potential Biological Importance. Z. S. Saify, M. Arif, S. Fozi, M. Shahabuddin and A. Rashid. *J. Pharm.*, University of Karachi, 5 (2): 29-132, June 1987
5. Effect of a Newly Synthesized Compound on Various Serum Lipid Parameters. M. Arif, Z. S. Saify, S. Rashid and M. Ahmed. *J. Islamic Acad. of Sci.*, 3(2): 118-123, June 1990.

6. Synthesis of some Phenacyl nicotines as Potential Anticholesterolmic Agent. M. Arif, Z. S. Saify and M. Ahmed, *Pak. J. Pram. Sci.*, 3(2): 63-67 July 1990
7. Effect of 2-methoxy phenacyl nicotinium bromide on Serum Total Cholesterol and Serum Transaminases. M. Arif, Z. S. Saify, A. B. Rehman, S. Rashid and M. Ahmed. *Journal of Islamic Academy of Sciences Turkey*, 4: 36-39 1991
8. Effect of Aspirin and Nicotinic acid Derivatives on Total Lipid and Phospholipid in Rabbits. M. Arif, Z. S. Saify, S. Asif Ali, A. B. Rehman, and S. Rashid. *Pakistan Journal of Cardiology*, 48-54, Oct., 1991.
9. Pharmacological effects of 3-methoxy Phenacyl nicotinium bromide Comparison with the Effect of Aspirin on Total Serum Cholesterol in Male Rabbits. M. Arif, Z. S. Saify, A. B. Rehman and M. Ahmed. *J. Islamic Academy of Sciences Turkey*. 5(2): 107-110, June 1992.
10. In vitro Antibacterial Studies of some Newly Synthesized Phenacyl Thiosemicarbazones. S. M. Khalid, S. A. Khan, M. Arif, Shoana Shafiq and Z. S. Saify. *Pak. J. Pharm. Sci.*, 5(2): 1992
11. Control of Seed-borne of Wheat by Microdilution Technique Utilizing Fusarium Oxysporum. R. Y. Hashmi, M. I. Khan, Z. S. Saify, and M. Arif *Proceedings of "Symposium on Status of plant pathology in Pakistan"*. 137-142, 1992.
12. Effect of Some Newly Synthesized Compounds and Comparison with the Clofibrate on total Serum Cholesterol in Experimentally Induced Hypercholesterolemia Rabbits. M. Arif, Z. S. Saify, A. B. Rehman and M. Ahmed. *Pak. J. of Pharmacol.*, 10(1) : 9-16 Jan., 1993.
13. Synthesis of Nicotinic Acid Derivatives Having Potential Therapeutic Activity. B. S. Siddiqui, Mansoor Ahmed, Sabira Begum, Z.S. Saify, M. Arif, A.H. Gilani and A. Latif. "The Proceedings of 19<sup>th</sup> IUPAC Symposium On "Chemistry of Natural Products" Karachi, Pakistan pp., 1-8, 1994.
14. Effect of New Derivatives of Nicotinic Acid on LDL Cholesterol and Comparison with Aspirin and Clofibrate as Anticholesterolmic Agent in Experimentally Induced Hypercholesterolemia white Male Rabbits. M. Arif, Z.S. Saify, A. B. Rehman. *J. Fac. Pharm. Ghazi*, 11(2), 151-166, Turkey.
15. In Vitro Antibacterial Studies of Four Newly Synthesized Phenacyl Thiosemicarbazone. Z. S. Saify, Shoana Shafique, M. Arif, S. M. Khalid and Shakeel Ahmed Khan. *Pak. J. Pharm. Sci.*, 9(2), July 1996.
16. Triterpenes, A Sterol and a Monocyclic Alcohol From *Momordica Carnatia*. Sabira Begum, Mansoor Ahmed, Bina S. Siddiqui, Abdullah Khan, Z. S. Saify and M. Arif *Phytochemistry*, 44(1), 1313-1320, 1997.
17. Pharmacological Evaluation of Newly Synthesized Piperidine Derivative for their Analgesic Activity. Rafeeq Alam Khan, Huma Rasheed, Z. S. Saify, Asif Bin Rehman and M. Arif *Pak. J. Pharm. Sci.*, 10(2), July, 17-20, 1997.

18. Effect of *Lachnanthes tinctoria* on certain Behavioral Parameters. Z. S. Saify, Fozia Noor, Nausheen Mushtaq, M. Arif and Shamoona Takween. Proceedings of First National Symposium on "Pharmacy, Pharmaceuticals and Health" Karachi, Pakistan 1997, pp. 211-215, 1997.
19. Biological Studies of Herbal Medicine, *Passiflora incarnata*. Z. S. Saify, Fozia Noor, Nausheen Mushtaq, M. Arif and Ahsana Dar. Proceedings of First National Symposium on "Pharmacy, Pharmaceuticals and Health" Karachi, Pakistan, pp. 223-231, 1997.
20. Effect of Alcoholic Extract of *Bryonia dioica* On the Spontaneous Activity of Isolated Rabbit Jejunum. Z. S. Saify, Nausheen Mushtaq, Fozia Noor, S. Naeem, M. Arif and Ahsana Dar. Published in the Proceedings of First National Symposium on "Pharmacy, Pharmaceutical and Health" Karachi, Pakistan, pp. 232-238, 1997.
21. Viscosity Studies on Mestranol. Shamim Akhtar, Tauqir Ahmed, Fauzia Hasan, M. Arif and Sohail Hasan. *J. Pharm. Sci.*, 11(2), July 1-4, 1998.
22. Platelet Activating factor Antagonists, Syeda Afroz, Azmat Ara, and M. Arif, *J. Pharm. Sci.*, 11(2), 5-12, 1998.
23. Role of Quinone moiety as anti tumor agent: A Review. Z S. Saify, Fozia Noor, Shamoona Takween and Mohammad Arif. *Pak. J. Pharm. Sci.*, 12(2), July 1999.
24. A Comparative Study of Omega-3-Fatty acids Obtained from Marine Fish and Bezafibrate Alone and in Combination as Hypolipidemic Agents. Z. S. Saify, Sohail Hassan, M. Arif, Faheem Ahmed and Kamran A. Chisti. *J. Pharm. Sci.*, 13 (1), 1-12, Jan. 2000.
25. Analgesic and Antimicrobial Activity of the Leaves Extracts of *Calendula Officinalis*. Z.S. Saify, Nausheen Mushtaq, Fozia Noor, M. Arif, Shamoona Takween and Shahida Perveen Ahmed. *Hamdard Medicus*. XLIII, No. 1, Jan.-March, 2000.
26. A Study on the Effects of New-Methyl Derivatives of Piperidine on Brain Catecholamine and Indolamine. Shamim Akhtar, Z.S. Saify, Hanifa Shehnaz, M. Arif and Darakhshan J. Haleem. Poster presented in 8<sup>th</sup> international Symposium on Natural product Chemistry 2000.
27. Comparative Studies On Nicotinic Acid Derivatives As Hypolipoproteinemic Agents Z.S. Saify, M. Arif, Shamim Akhtar, Nausheen Mushtaq, and Farah Mazhar. *Pak. J. Pharm. Sci.*, Vol. 14(2), 33-42, 2001.
28. A Study on the fatty acid composition of fish liver oil from two marine fish, *Eusphyra blochii* and *Carcharhinus bleekeri* Z.S. Saify, Shamim Akhtar, Khalid Mohammad Khan, Shehnaz Perveen, S.A.M. Ayattollahi, Sohail Hassan, M. Arif, S.M. Haider, Faheem Ahmad, Sonia Siddiqui and Zarar Khan. *Turk. J. Chem.* 27, (2003), 251-258.
29. Biochemical studies on Marine Fish oil Part-I: Effects of fish oil and lipid lowering drugs on LDL /HDL cholesterol levels. Z.S. Saify, Faheem Ahmed, Sonia Siddiqui, Shamim Akhtar, Sohail Hasan, M. Arif, Shaheen A. Hussain and Nausheen Mushtaq. *Pak. J. Pharm. Sci.* vol. 16 (2) 2003, 1-8.
30. Effects of fish oil and lipid lowering drugs on serum triglycerides alone and in

- combination. Z.S. Saify, Faheem Ahmed, Shamim Akhtar, Nousheen Mushtaq, M. Arif and Sonia Siddiqui. *Pak. J. Pharmacology*. Vol. 21(2), 2004, 1-7
31. Antibacterial Activity of 1-methyl-7-methoxy- $\beta$ -carboline and its phenacyl and coumarine analogues. Z.S. Saify, Jawaid Farhad, Nousheen Mushtaq, Fozia Noor, Shamim Akhtar Baqir Shyum Naqvi, Muhammad Arif, Haris Shoaib and Khalid M. Khan. *Pak. J. Pharm. Sci.* vol. 18 (1), 2005.
  32. Neurochemical Estimation of Some New Quaternary Phenacyl Bromo Piperidinium Compounds. Z.S. Saify, Shamim Akhtar, M. Arif, Nousheen Mushtaq and D. J. Haleem. *Pak. J. Pharm. Sci.* vol. 18 (2), 2005.
  33. Biochemical Studies of Marine Fish Oil, Part II: Effects Of Fish Oil and Lipid Lowering Drugs On Total Cholesterol And Total Lipid. Z.S. Saify<sup>+</sup>, Fahim Ahmed, Shamim Akhtar and M. Arif. *Pak. J. Pharmacology*, 22(1), 2005.
  34. Synthesis and Evaluation of novel phenacyl based carboxamide derivatives Piperidine as antihypertensive agents. Shamim Akhtar, Z. S. Saify, M. Arif, Nousheen Mushtaq, Ahsana Dar and Ashfaq Ahmed (*Pakistan Journal of Physiology*, 2(1) pp; 42-48, 2006
  35. Mushtaq, N., Saify, Z. S., Noor, F., Takween, S., Akhtar, S., Arif, M. and Khan, K. M. Synthesis And Behavioral study of 4-(1-Pyrrolidinyl) piperidine and its Derivatives. *Pak. J. Pharmacol. Sci* vol. 24 (2), 2007, pp. 1-6
  36. Mushtaq, N., Saify, Z. S., Noor, F., Takween, S., Akhtar, S., Arif, M. and Khan, K. M. Synthesis and Pharmacological Activities Of 7-Azaindole Derivatives (accepted for publication in *Pak. J. Pharm. Sci* vol. 21, 2008).